Skills Summary

Estimating and Solving with Decimals

Use this reference while completing your worksheet or when studying.

Skill 1 — Estimating by rounding first

Rule: to estimate, round *every* number first and then compute. Do not work out the exact answer and round it at the end — that is checking, not estimating.

How to round: look at the digit one place to the right of the place you are rounding to. If it is 5 or more, round up; if it is 4 or less, round down.

Why bother: the estimate tells you roughly where the exact answer must land, so a misplaced decimal point stands out at once.

Example: Estimate $27.4 + 41.8$ by rounding each number to the nearest ten.
Step 1. The question says estimate to the nearest ten, so I round both numbers to the nearest ten before I add anything.
Step 2. 27.4 is nearer 30 than 20, and 41.8 is nearer 40 than 50: $30 + 40 = 70$. $\Rightarrow$ **70**

Try it: Estimate $38.2 + 24.9$ by rounding each number to the nearest ten.

check: **60**

Skill 2 — Adding and subtracting decimals

Rule: line up the decimal points, one under the other, and fill any short number with zeros on the end — 0.4 and 0.40 are the same amount.

Two-step money problems: add everything that was spent first, then subtract that one total from the money you started with. Write a whole dollar amount as 50.00 so the columns line up.

Example: Ben has \$20. He spends \$7.45 and \$3.80. How much is left?
Step 1. "How much is left" means subtracting, but two things were bought — so I add the two prices into one total first, then take that away from \$20.
Step 2. $7.45 + 3.80 = 11.25$, and then $20.00 - 11.25 = 8.75$. $\Rightarrow$ **8.75**

Try it: Ana has \$30. She spends \$12.60 and \$4.75. How much is left?

check: **12.65**

Skill 3 — Multiplying and dividing decimals

Multiplying: ignore the decimal points and multiply the digits, then count how many digits sat after the decimal points altogether and put that many after the point in your answer.

Dividing by a whole number: divide as usual and keep the decimal point in line.

Dividing by a decimal: make both numbers ten (or a hundred) times bigger first — $4.2 \div 0.6$ is the same as $42 \div 6$.

Example: Find 4×3.25, and then $9.6 \div 4$.
Step 1. Both are exact calculations, so I multiply the digits and count decimal places for the first, and share into equal parts for the second.
Step 2. $4 \times 325 = 1300$ with two decimal places gives 13.00; and $96 \div 4 = 24$, so $9.6 \div 4 = 2.4$.
$\Rightarrow$ **13** and **2.4**

Try it: Find 5×1.8, and then $7.5 \div 5$.

check: **9** and **1.5**

Skill 4 — Comparing and ordering decimals

Rule: give every number the same number of decimal places by writing zeros on the end, then compare them place by place from the left.

Careful: a longer decimal is not automatically bigger — 0.58 has more digits than 0.6 but is smaller, because $0.6 = 0.60$ and $60 > 58$.

Example: Which is greater, 0.6 or 0.58?
Step 1. The two numbers have different numbers of decimal places, so before comparing anything I give them the same number by writing a zero on the end.
Step 2. 0.6 becomes 0.60; comparing hundredths, 60 beats 58. $\Rightarrow$ $\mathbf{0.6 > 0.58}$

Try it: Write $<$ or $>$ between 2.05 and 2.5.

check: $\mathbf{2.05 < 2.5}$